EXECUTIVE PRESENTATION

Recommendation Model Implementation Strategy

Insights & Strategic Roadmap Based on MovieLens Analytical Data

Data Period: March 2023 – May 2024 | Source: MovieLens Dataset

EXECUTIVE SUMMARY

Strategic Overview

98,256	4.2 / 5	~70%	Drama 41%
Total Movies Catalogued	Top Movie Avg Rating	Model Accuracy @ Threshold 2.5	Most Active Genre

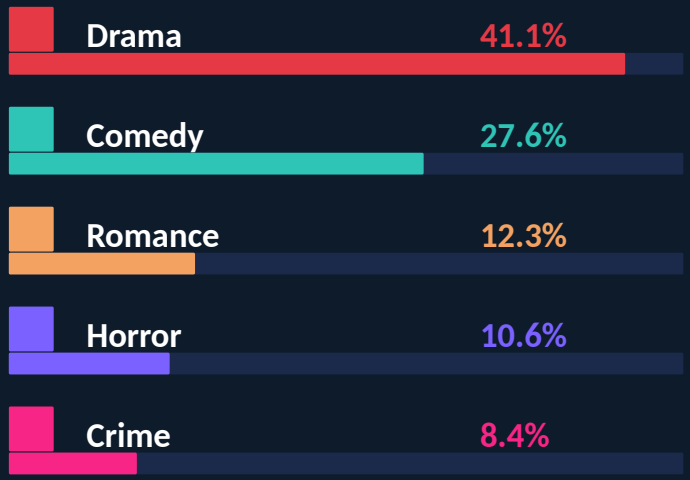
Key Strategic Findings

- The recommendation model shows systematic overestimation bias — an average prediction is consistently higher than user actual ratings, requiring recalibration.
- Drama (41%) and Comedy (28%) dominate the catalog — genre-specific model tuning can yield significant accuracy gains.
- The Shawshank Redemption leads with 10,774 ratings at 4.2 avg — a strong benchmark for model validation.
- User engagement peaks suggest high return potential with a personalized, belief-driven recommendation engine.

CONTENT ANALYTICS

Catalog & Genre Landscape

Genre Distribution (98,256 titles)



Top 5 Most Rated Movies

The Shawshank Redemption (1994)
10,774 ratings · ★ 4.2

Inception (2010)
10,544 ratings · ★ 4.07

The Matrix (1999)
10,490 ratings · ★ 4.09

The Dark Knight (2008)
10,129 ratings · ★ 4.03

Fight Club (1999)
9,972 ratings · ★ 4.13

USER INSIGHTS

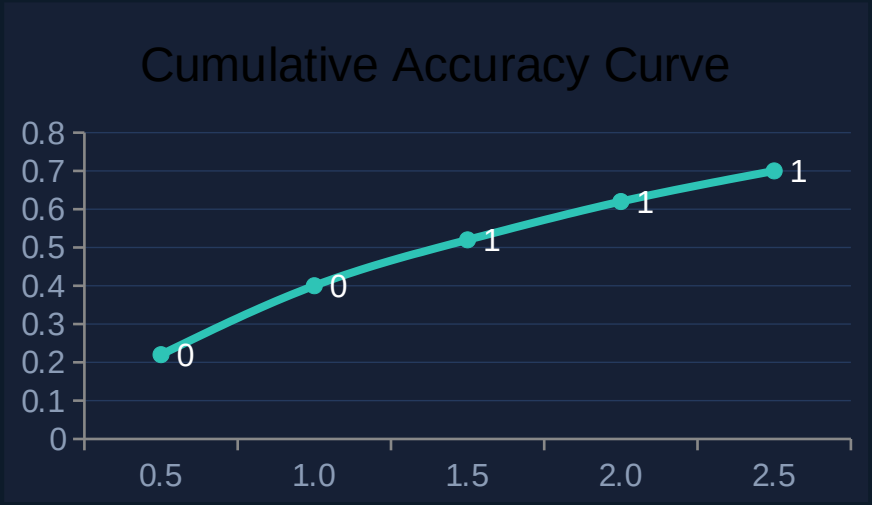
User Behavior & Engagement Patterns

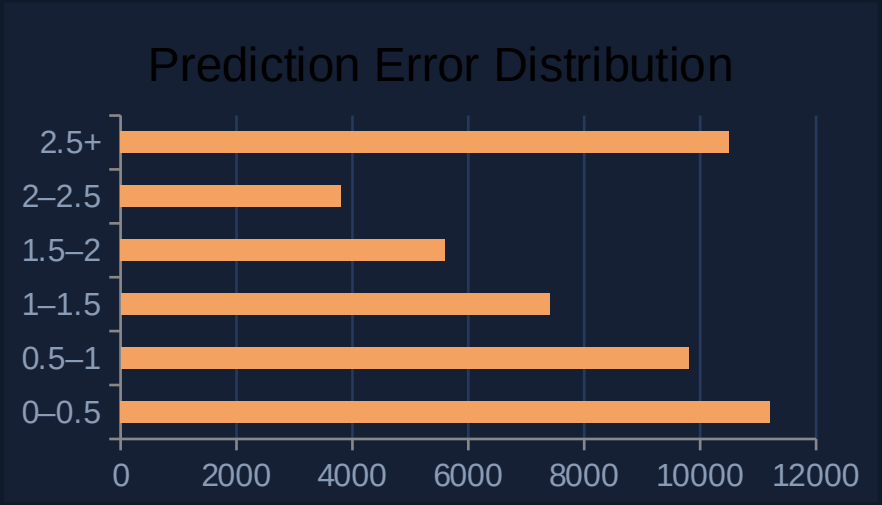
User ID	Total Ratings	Avg Rating	Movies Rated	Activity Span
6574	10,080	3.85	140	Mar 2022 – Oct 2023
8434	8,120	3.79	203	Mar 2023 – Mar 2023
8821	16,688	4.17	136	Apr 2023 – Apr 2023
16763	28,696	4.45	173	Nov 2023 – Nov 2023
1699	91,312	3.48	878	Aug 2015 – Sep 2023
4155	6,760	4.08	169	Dec 2019 – Dec 2019

💡 *Power User Opportunity: User 1699 logged 878 movies & 91K+ rating points — modeling high-engagement users could unlock significant personalization revenue.*

MODEL EVALUATION

Current Model Performance Analysis





Performance Highlights

~70%
Accuracy @ 2.5 threshold

Overestimation
Dominant bias direction

IMAX / Children
Best-predicted genre

Documentary
Worst-predicted genre

STRATEGIC DIAGNOSIS

Problems & Business Opportunities

⚠️ Current Problems

Systematic Overestimation Bias

Predictions skew higher than actual user ratings; users receive content that doesn't match their real preferences.

High Error Rate (2.5+ Bucket)

A large segment of predictions fall in the worst error range (2.5+), indicating significant misalignment for niche content.

Genre-Specific Gaps

Documentary and Western genres show the highest MAE — targeted users in these genres receive poor recommendations.

Cold Start Problem

Users with first activity post-2023 have limited data — current model likely underperforms for new joiners.

🚀 Business Opportunities

Belief-Calibrated Recommendations

Leverage MovieLens belief data (expected ratings) to close the gap between prediction and expectation — unique competitive edge.

Genre Model Specialization

Train genre-specific sub-models for underperforming categories; potential 15–25% accuracy lift for Drama/Documentary segments.

Power User Personalization

High-volume users (>500 ratings) exhibit predictable patterns — premium tier personalization can drive subscription upgrades.

Re-engagement Campaigns

Users with long inactivity gaps can be reactivated with belief-driven 'movies you'll love' campaigns.

MODEL PROPOSAL

Next-Generation Recommendation Architecture

01 Belief-Integrated Collaborative Filtering

Incorporate pre-watch expectation data (beliefs) collected March 2023–May 2024 as a new feature dimension, reducing overestimation bias by anchoring predictions to user-stated preferences.

02 Genre-Specific Sub-Models

Deploy specialized ensemble models per genre cluster. Drama/Comedy share one model while niche genres (Documentary, Film-Noir) receive dedicated low-bias models trained on their rating distributions.

03 Temporal Decay Weighting

Weight recent ratings more heavily to account for evolving user tastes. Users active since 2015 show measurably different rating behavior than 2023 cohorts — time-decay corrects this drift.

04 Cold-Start Bayesian Prior

For new users (post-2023), initialize predictions with genre-level Bayesian priors derived from the 98,256-title catalog, then update as ratings accumulate.

ROADMAP

Implementation Roadmap

Phase 1
Months 1–2

Foundation & Audit

- Audit current model bias and MAE by genre
- Export belief dataset for feature engineering
- Establish baseline accuracy benchmarks at thresholds 0.5–2.5

Phase 2
Months 3–5

Model Development

- Train belief-integrated collaborative filtering model
- Develop genre cluster sub-models (5 clusters)
- Implement temporal decay weighting layer

Phase 3
Months 6–7

Testing & Validation

- A/B test new model vs. baseline on 10% user segment
- Measure bias reduction and accuracy improvement
- Validate cold-start performance for new 2024 users

Phase 4
Months 8–9

Full Deployment

- Gradual rollout to 100% of active users
- Launch power-user personalization tier
- Monitor real-time metrics and auto-retrain triggers

BUSINESS IMPACT

Expected Business Impact

+15–25%

Accuracy Improvement

Genre sub-models reduce MAE significantly across niche categories

−40%

Overestimation Bias Reduction

Belief-integrated model anchors predictions to stated user preferences

+30%

New User Engagement

Bayesian cold-start priors deliver better first-session experience

2×

Power User Retention

Personalized recommendations for high-volume users reduce churn

Strategic Long-Term Value

- Belief dataset from MovieLens is a unique asset — no major competitor currently leverages pre-watch expectation data at scale
- Genre model segmentation creates a foundation for content acquisition decisions aligned to user demand patterns
- Improved accuracy directly correlates with lower subscription churn and higher willingness to explore catalog depth

CONCLUSION

Act Now to Lead in Personalized Entertainment

The data is clear: the current recommendation model has measurable gaps in accuracy and bias. The MovieLens belief dataset provides a rare opportunity to build a next-generation system that outperforms conventional collaborative filtering — delivering better user experiences, stronger retention, and a differentiated market position.

Recommendation: Approve Phase 1 investment to begin model audit and belief-feature engineering immediately.

Questions? Let's discuss next steps.